



SPONTANEOUSLY ARISING DISEASE: REVIEW ARTICLE

Feline Idiopathic Cystitis: Pathogenesis, Histopathology and Comparative Potential

Emily Jones^{*}, Chiara Palmieri^{*}, Mary Thompson[†], Karen Jackson^{*}
and Rachel Allavena^{*}

^{*} School of Veterinary Science, The University of Queensland, Gatton, Queensland and [†] School of Veterinary and Life Sciences, Murdoch University, Murdoch, Western Australia, Australia

Summary

Bladder pain syndrome (BPS) is a debilitating disease in humans, particularly women, with patients experiencing chronic, intractable, lower urinary and pelvic pain. Although rodent models have been used, feline idiopathic cystitis (FIC) is a naturally occurring bladder disease of cats that is frequently considered to be the preferred model for BPS. Histologically, FIC is most similar to the non-Hunner BPS subtype. Histology is unnecessary for the clinical diagnosis of FIC but is of great value in elucidating the pathogenesis of this disease so that prevention and therapeutic interventions can be optimized. Further study of the histological features of FIC and BPS is required to determine the significance of Von Brunn's nests, which are invaginations of hyperplastic urothelium that have been associated with irritative bladder stimuli in animals and have been observed in FIC. We review the possible pathogenesis, histopathological similarities and differences between FIC and BPS, and highlight the potential of FIC as a model of BPS.

© 2021 Elsevier Ltd. All rights reserved.

Keywords: animal model; bladder pain syndrome; feline idiopathic cystitis; histopathology

Contents

Introduction	19
Clinical Signs and Epidemiology	19
Risk Factors	19
Subtypes of BPS and FIC	20
Pathogenesis	20
The Role of Histopathology in Diagnosis of BPS and FIC	22
Von Brunn's Nests	24
Conclusions	25
CRediT Author Statement	25
Acknowledgments	25
Conflict of Interest Statement	25
Funding	25
References	25

Introduction

Feline idiopathic cystitis (FIC) is a spontaneously occurring, non-infectious bladder disease of cats, while bladder pain syndrome (BPS), previously termed interstitial cystitis, is a chronic, waxing and waning lower urinary tract and pelvic disease of humans (van de Merwe and Arendsen, 2000; van de Merwe *et al.*, 2008; Tirumuru *et al.*, 2010). As similar pathogenic mechanisms have been proposed for FIC and BPS, FIC is considered to be the ideal disease model for BPS (Roth *et al.*, 1990; Jabr and Fry, 2013; Lascelles *et al.*, 2015). There are two subtypes of BPS, with different histological features (Maeda *et al.*, 2015; Kim *et al.*, 2017), but these features are not well-defined and have not been rigorously compared with the histological changes observed in FIC. The terms idiopathic cystitis and interstitial cystitis have been used interchangeably in both the human and veterinary literature. In this review, the human disease is referred to as BPS while FIC will be used to denote the feline condition.

Clinical Signs and Epidemiology

The nomenclature and diagnosis of FIC has long been a source of confusion for veterinarians and veterinary pathologists. The definition of FIC is inconsistent throughout the literature, particularly with regard to the presence of urethral plugs and whether their presence should preclude a diagnosis of FIC. The most commonly accepted clinical definition of FIC is the presence of chronic, waxing and waning clinical signs of irritative voiding (pollakiuria, stranguria and dysuria) with an absence of neoplasia or bacteriuria, with or without the presence of uroliths or urethral plugs (Kruger *et al.*, 1991, 2009; Defauw *et al.*, 2010; Lulich *et al.*, 2010). Uroliths and urethral plugs are thought to be sequelae of underlying urinary bladder or metabolic abnormalities and are not specific to FIC (Osborne *et al.*, 1996b).

The welfare of cats with FIC is severely compromised, with a mortality rate of 12.5% that increases to 26% in cats with obstructive FIC, primarily due to elective euthanasia following repeated obstructive or non-obstructive episodes (Gerber *et al.*, 2008; Defauw *et al.*, 2011). The prevalence of feline lower urinary tract signs in the general cat population is 2–4.6% (Lund *et al.*, 1999, 2012; Kirk *et al.*, 2001; Ikeda *et al.*, 2009), with around two thirds of these cases attributed to FIC (Lekcharoensuk *et al.*, 2001; Cameron *et al.*, 2004; Defauw *et al.*, 2011; Lemberger *et al.*, 2011; Dorsch *et al.*, 2014). Therefore, approximately one in every 200 cats may require euthanasia

due to the current lack of knowledge regarding the pathogenesis of FIC.

Cats with FIC may present with varying combinations of clinical signs, including dysuria (77%), stranguria and periuria (70%), pollakiuria (78%) and macroscopic haematuria (71%) (Bufington, 2011; Defauw *et al.*, 2011). Episode duration for FIC is 2–90 days (median 6.5 days), with 78% of cats experiencing more than one episode and 50% having a mean inter-episode interval of less than 6 months (Defauw *et al.*, 2011). Male cats are predisposed to potentially fatal urethral obstruction due to their narrow urethra (Osborne *et al.*, 1996a). Sixty-seven per cent of obstructed cats have crystalluria without urolithiasis, suggesting that the obstruction was due to causes other than uroliths, including protein or mucus plugs, or urethral spasm (Defauw *et al.*, 2011).

On the other hand, BPS is currently defined as chronic, debilitating, hyperaemic inflammation and pain of the urinary bladder of more than 6 months' duration, in the absence of any other bladder pathology (Lowe *et al.*, 1997; Denson *et al.*, 2000; van de Merwe *et al.*, 2008; Grover *et al.*, 2011). Pain on bladder filling is a defining feature of BPS (van de Merwe *et al.*, 2008; Davis *et al.*, 2014), along with other symptoms such as nocturia, urinary urgency and increased frequency (Davis *et al.*, 2014). The two BPS appears to cause similar symptoms, but differ in severity and histological features (Kabjorn-Gustafsson and Peeker, 2013). There is wide variation in clinical presentation, quality of life and cystoscopic and biopsy findings in BPS, which has caused inconsistency in diagnosis, treatment and prognosis (van de Merwe *et al.*, 2008).

The prevalence of BPS in women is thought to be around 0.5%, although it may be as high as 12% (Rosenberg and Hazzard, 2005), with up to 0.04% of men affected (Grover *et al.*, 2011; Davis *et al.*, 2014; Arora and Shoskes, 2015). Prevalence increases from 0.5% up to 1.4% in women who have a close relative with BPS, suggesting a potential hereditary component to the disease (Warren *et al.*, 2004; Davis *et al.*, 2014).

Risk Factors

Risk factors for FIC include higher body weight, younger age (<10 years) (Chew *et al.*, 2011a), higher number of cats in the household, lower activity level, less hunting behaviour, higher body condition score, lower water intake, using a litter box and having less access to the exterior environment (Defauw *et al.*, 2011; Houbrechts, 2018). Magnesium ammonium phosphate (struvite) crystals have been found in the urine of up to 58% of cats with FIC (Defauw *et al.*,

2011; Kennedy *et al*, 2016). However, it is generally accepted that crystals are commonly present in feline urine, with crystalluria documented in 41/99 (41%) healthy, middle-aged to older cats during health screening (DiBartola, 2010; Paepe *et al*, 2013). Furthermore, as struvite is the mineral most frequently detected in feline urethral plugs, it may play a pathogenetic role in FIC, although information is limited (Osborne *et al*, 2009; Okafor *et al*, 2019). Currently, urinary crystal formation is thought to be exacerbated by the inflammation caused by FIC, but it is not believed to be involved in the primary pathogenesis of FIC (Chew *et al*, 2011a). In summary, cats with FIC are most likely to be overweight, male, young adults that live indoors.

In contrast to FIC, neither crystalluria nor urolithiasis appear to be features of BPS. Many individuals with BPS experience non-urinary tract problems such as other chronic pain disorders (Janicki, 2003), endometriosis (Tirlapur *et al*, 2013), psychiatric disorders (Malykhina and Hanno, 2014), drug hypersensitivity (Peeker *et al*, 2000), irritable bowel syndrome or asthma or other allergies (Warren *et al*, 2011). It has been hypothesized that both BPS and FIC are part of a broader, systemic problem involving the brain, nerves and other organ systems (Buffington, 2004, 2011a; Keller *et al*, 2012). Adverse events occurring early in life, including physical, emotional or sexual abuse, have been linked to chronic pain disorders and are frequently identified in BPS patients, prompting the description of BPS as a functional somatic syndrome (Warren, 2014; Dinis *et al*, 2015). It has also been hypothesized that FIC development could reflect a combination of genetics and fetal or early life stressors, similar to some chronic human syndromes, including BPS (Buffington, 2009). These factors are thought to result in an overactive sympathetic nervous system, leading to FIC as well as comorbidities such as intestinal, behavioural, dermatological, epithelial, neurological, endocrine or immune disorders (Buffington, 2011a). Naturally occurring FIC represents a more comparable bladder disease to BPS than any other inflammatory bladder disease model (Westropp and Buffington, 2002).

Subtypes of BPS and FIC

It has long been suggested that there are two subtypes of BPS: Hunner-type BPS (HBPS), previously termed classic or ulcerative BPS, and non-Hunner (NHBPS) non-ulcerative BPS, (Grover *et al*, 2011). Hunner's lesions are distinct inflammatory lesions involving rupture of the bladder mucosa and submucosa due to increasing distension of the bladder, and have been observed during cystoscopy (Hunner, 1915;

van de Merwe *et al*, 2008). Hunner's lesions appear cystoscopically as erythematous, well-circumscribed mucosal areas with small vessels radiating towards a central scar (van de Merwe *et al*, 2008; Lucon *et al*, 2010).

It has been hypothesized that HBPS and NHBPS could be distinct diseases (Leiby *et al*, 2007; Peters *et al*, 2011), a concept that was recently consolidated by an expert panel (Whitmore *et al*, 2019). The confusion over the two BPS subtypes frequently complicates the literature on these diseases. For example, Erickson *et al* (2008) evaluated 'Hunner-type cases' but included patients with 'glomerulations or ulcers'. Glomerulations, defined as punctate submucosal petechial haemorrhages, can be found in both BPS subtypes but are considered a feature of NHBPS (Kiuchi *et al*, 2009). Glomerulations have been included in the diagnostic criteria for FIC (Westropp, 2003) but, more recently, were found not to be significantly associated with BPS (Wennevik *et al*, 2016), questioning the definition of NHBPS. For this reason, it would be prudent to diagnose all BPS patients using cystoscopy and histopathology as per the European Society for the Study of Interstitial Cystitis (ESSIC) guidelines, and stratify patients by Hunner subtype when investigating disease biomarkers or treatment modalities. It is important to note that many of the studies evaluated for this review, particularly the older ones, did not or were unable to differentiate the two BPS subtypes. Therefore, any significant changes may have been attributable to both subtypes combined, or to only one subtype, which would have skewed the results. Despite this inconsistency in human case definitions, FIC is widely supported as a model for BPS, particularly NHBPS (Buffington *et al*, 1999; Jabr and Fry, 2013; Malykhina and Hanno, 2014).

Pathogenesis

The precise etiology and pathogenesis of BPS and FIC are unknown but are hypothesized to involve multiple factors including increased bladder wall permeability, neurological abnormalities, stress, environmental and genetic factors, and concurrent diseases such as autoimmune, infectious or systemic disease processes.

One of the major pathogenetic factors in BPS and FIC is increased bladder wall permeability, which enables components of the urine, particularly potassium, to access deeper layers of the bladder wall and stimulate sensory nerves (Chew and Buffington, 2013; Dyer and Twiss, 2014). One mechanism of increased bladder wall permeability is compromised urothelium. BPS and FIC bladder tissue have reduced transepithelial

resistance, increased water and urea permeability and altered ATP release across the urothelium, compared with controls (Birder *et al.*, 2003; Izgi and Daneshgari, 2013; Davis *et al.*, 2014). Urine concentrations of nitric oxide (NO), an inflammatory mediator, are elevated in patients with BPS and appear to facilitate differentiation between HBPS and NHBPS (Logadottir *et al.*, 2004; Birder, 2014; Davis *et al.*, 2014). NO can be synthesized and released by nerve and urothelial cells within the bladder wall, and elevated urinary concentrations can cause increased permeability of the urothelium to water, urea and other substances (Birder, 2014). NO has also been found to be elevated in FIC cats compared with healthy controls (Birder *et al.*, 2005). Laser removal of damaged urothelium has been associated with reduced pain symptoms in HBPS (Birder, 2014). Another facet of urothelial compromise is an altered urothelial glycosaminoglycan (GAG) layer (Lucon *et al.*, 2010; Chew *et al.*, 2011a; Davis *et al.*, 2014). The GAG layer is a luminal layer of mucin comprised of sulphonated GAGs and glycoproteins, which plays an important role in preventing bacterial adherence and otherwise providing physical protection to the urothelium (Birder, 2014). Increased concentrations of free uronic acid, a common GAG component, as well as sulfonated GAGs in the urine, have been positively associated with the severity of BPS symptoms in humans (Lokeshwar *et al.*, 2005; Izgi and Daneshgari, 2013). Abnormal protein and chondroitin sulphate expression patterns were observed in the urothelium of 7/8 FIC cats, while only 2/7 normal feline bladders showed slight changes in expression patterns (Hauser *et al.*, 2015). It is currently unclear whether these urothelial changes are causative or secondary to the disease. Furthermore, bladder wall inflammation causes leakage of serum proteins into the urine, resulting in precipitation of these proteins and elevation of urine pH, contributing to struvite crystal precipitation and feline urethral plug formation (Westropp *et al.*, 2005; Defauw *et al.*, 2011). Finally, trophic factors such as heparin-binding epidermal growth factor-like factor have been associated with abnormalities in urothelial growth and proliferation in humans (Birder, 2014).

Neurological abnormalities are a second hypothesized feature of BPS and FIC. One pathogenic mechanism involves neurogenic inflammation, with increased C-fibre neuron sensitivity, and upregulation of the stress response system (SRS), indicating increased sympathetic activation (Chew and Buffington, 2013) due to environmental stressors (Westropp and Buffington, 2002; Westropp, 2003). SRS upregulation is believed to be responsible for the increased acoustic startle reflex seen in one study in FIC cats without environmental enrichment

(Hague *et al.*, 2013). Increased immunoreactivity for substance P, a neurokinin neurotransmitter, and increased density of neurokinin-1 receptors have been associated with BPS and FIC (Buffington and Wolfe, 1998; Moldwin, 2002; Izgi and Daneshgari, 2013). Furthermore, larger dorsal root ganglion cell bodies and abnormal neuropeptide profiles and hyperexcitability have been found in FIC cats (Sculptoreanu *et al.*, 2005), suggesting a role for anatomical and physiological changes in sensory neurons. Finally, altered acetylcholine synthesis and release in the oesophageal mucosa have been identified in FIC cats, suggesting that systemic cholinergic changes may be involved in altering systemic sensory function and visceral hyperalgesia (Birder *et al.*, 2011). Neurogenic changes, such as upregulation of neuro-inflammatory mediators, have been observed in BPS patients compared with controls (Sant and Theoharides, 1994). Nerve growth factor (NGF), a neurotrophic factor, has been found to be elevated in BPS patients compared with controls, but NGF has not been evaluated in FIC (Lowe *et al.*, 1997; Jhang and Kuo, 2016).

Stress has long been postulated as a risk factor for the development of BPS (Davis *et al.*, 2014) and FIC (Amat *et al.*, 2016). A major hypothesis in FIC research relates to abnormalities of the hypothalamic–pituitary–adrenal axis. Cats diagnosed with FIC have significantly ($P < 0.05$) smaller adrenal gland weight and volume and a reduced response to adrenocorticotrophic hormone stimulation compared with controls, suggesting that a primary adrenal insufficiency may be associated with FIC (Buffington and Pacak, 2001; Westropp *et al.*, 2003; Nicolle, 2006). One proposed cause of small adrenal glands is maternal perception of environmental threats affecting fetal adrenal gland development and potentially affecting the SRS (Buffington, 2009; Chew and Buffington, 2013). In humans, lower overall stress levels have been associated with reduced pain and urgency in BPS patients (Davis *et al.*, 2014). Furthermore, increased activation of the sympathetic nervous system may lead to altered bladder permeability, as assessed by measuring bladder noradrenaline concentration and adrenoceptor function (Buffington *et al.*, 2002) and increased plasma noradrenaline activity in cats with FIC compared with controls (Buffington and Pacak, 2001).

In addition to physiological factors, there have been suggestions of a genetic predisposition to BPS and FIC. First-degree female relatives of BPS patients were 17 times more likely to have BPS than any other female in the general population (Warren *et al.*, 2004). There is a strong concordance among monozygotic twins both having BPS symptoms (55%; $n = 9$),

with no such shared symptoms in dizygotic twins ($n = 25$) (Grover *et al*, 2011). In cats, there seems to be a higher incidence of FIC in pedigree animals compared with mixed breeds (Defauw *et al*, 2011).

Various systemic diseases have been associated with BPS and FIC. Hypotheses exist around the involvement of pelvic floor disorders and chronic pain disorders in BPS patients (Peters and Carrico, 2006; Warren *et al*, 2011; Keller *et al*, 2012). The presence of an extraordinary component of BPS was further suggested by one small study in which pelvic pain persisted after removal of the bladder, uterus, ovaries and fallopian tubes in four patients (Baskin and Tanagho, 1992). An autoimmune component to BPS was postulated after antinuclear antibodies were found to be increased in the bladder of BPS patients compared with controls that had normal bladders and patients with bacterial cystitis (Davis *et al*, 2014). ‘Non-infectious cystitis’, including BPS, has been associated with Crohn’s disease (Grover *et al*, 2011) and a more recent publication suggests that subclinical infections, ageing, autoimmunity and sterile inflammation may initiate a chronic inflammatory response, leading to mood disorders as well as to chronic inflammatory diseases such as BPS (Fleshner *et al*, 2017).

Increased urothelial gamma delta T cells and increased urothelial and submucosal IgA and IgM plasma cells have been identified in BPS compared with cases of bacterial cystitis and healthy controls (Christmas, 1994). Mast cell degranulation contributes to the inflammation observed in BPS patients (Erickson *et al*, 2008; Grover *et al*, 2011; Yoshimura *et al*, 2014) and has also been implicated in the pathogenesis of FIC (Gunn-Moore, 2002; Mohamaden *et al*, 2019).

The Role of Histopathology in Diagnosis of BPS and FIC

Bladder biopsy is currently not routinely used as a diagnostic tool for BPS or FIC. Histopathological findings in BPS have been variable, but generally include non-specific changes associated with chronic inflammation, in addition to the possible and controversial theory of mast cell involvement (van de Merwe *et al*, 2008; Fariello and Whitmore, 2013). However, histology does have a role as an adjunct tool to determine BPS severity and provide information on prognosis and management, as well as to exclude underlying neoplasia. Histological changes observed in BPS and FIC are summarized in Table 1 and Figs. 1 and 2. BPS patients, with or without Hunner’s lesions, have histological changes of

Table 1
Histological comparison of human bladder pain syndrome and feline idiopathic cystitis

<i>Histological changes</i>	<i>HBPS</i>	<i>NHBPS</i>	<i>FIC</i>
Urothelial denudation or ulceration ^{1,2,3,4}	Yes, 6% remaining	Yes, 70% remaining	Yes
Submucosal oedema ^{1,5,6,7}	Yes	Yes	Yes
Chronic inflammatory cell infiltrates ^{1,2,3,4,6,8,9}	Yes, more severe than NHBPS ¹⁰	Yes	Yes, moderate
Granulation tissue in submucosa ^{2,7,11,12}	Yes	Yes	NR
Increased numbers of mast cells in submucosa ^{1,2,7,8,9,13}	Yes	Yes, mild	Yes*
Muscularis fibrosis ^{1,2,9,12}	Yes	Yes, more severe than Hunner-type disease ¹⁰	Yes
Submucosal blood vessel dilation, ¹ haemorrhage ^{1,5} or vascular reaction ⁵	NR	NR	Yes
Increased suburothelial proliferation in the form of Von Brunn’s nests ⁸	NR	NR	Yes

* One study reported an increase in only degranulated submucosal mast cells.

HBPS, Hunner-type bladder pain syndrome; NHBPS, non-Hunner bladder pain syndrome; NR, not reported.

¹Chew *et al*, 2011b; ²Kabjorn-Gustafsson and Pecker, 2013; ³Lowe *et al*, 1997; ⁴Lynes *et al*, 1990; ⁵Buffington *et al*, 1999; ⁶Srikousthubha *et al*, 2013; ⁷Tomaszewski *et al*, 2001; ⁸Kullmann *et al*, 2018; ⁹Specht *et al*, 2004; ¹⁰Kim *et al*, 2017; ¹¹Hunner, 1915; ¹²van de Merwe *et al*, 2008; ¹³Mohamaden *et al*, 2019.

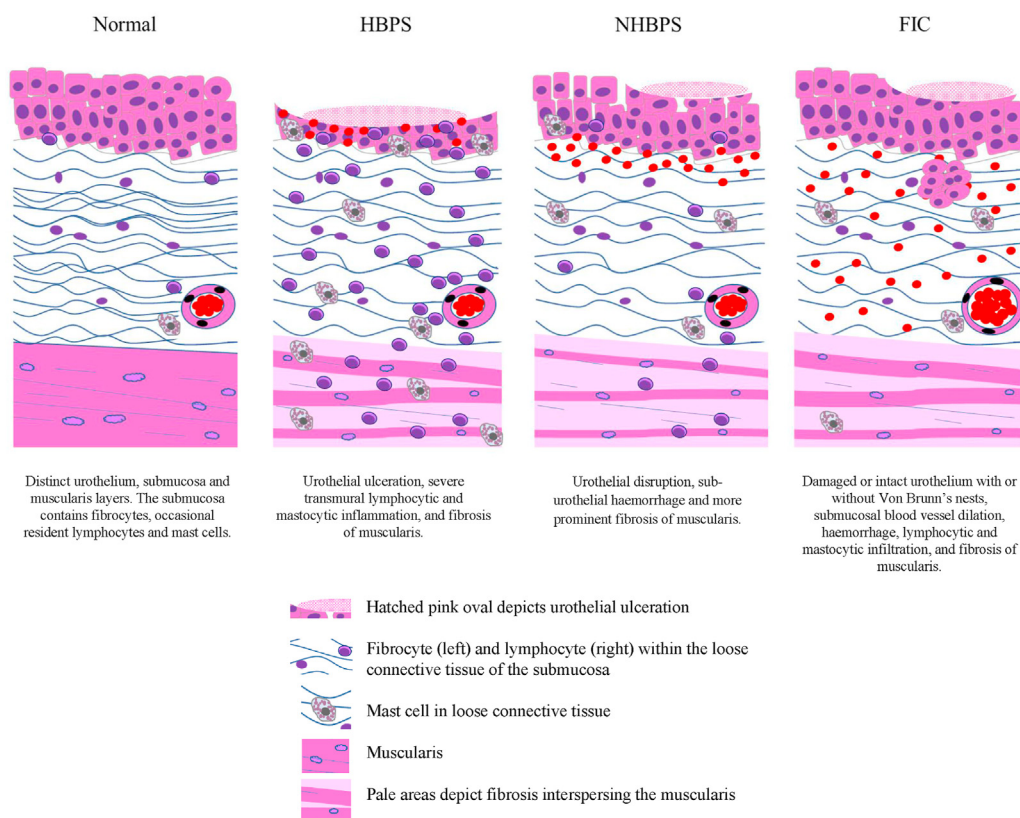


Fig. 1. Diagrammatic depiction of normal and BPS/FIC disease states. Normal bladder contains distinct urothelium, submucosa and muscularis layers. The submucosa contains fibrocytes, occasional resident lymphocytes and mast cells. Histological features of HBPS include urothelial ulceration, severe transmural lymphocytic and mastocytic inflammation, and fibrosis of the muscularis. Histological features of NHBPS include urothelial disruption, suburothelial haemorrhage and more prominent fibrosis of muscularis compared with HBPS. FIC histological features include damaged or intact urothelium with or without Von Brunn's nests, submucosal blood vessel dilation, haemorrhage, lymphocytic and mastocytic infiltration, and fibrosis of the muscularis.

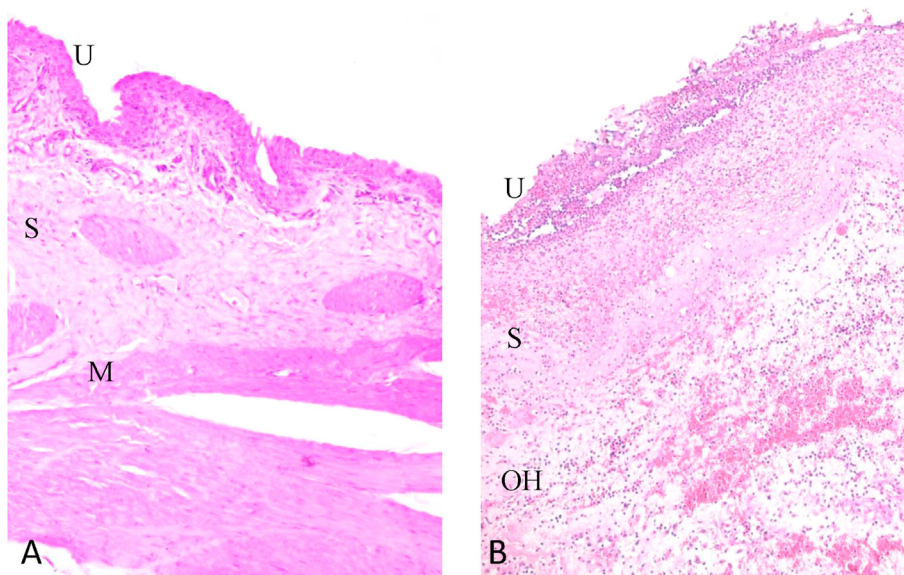


Fig. 2. (A) Cat, urinary bladder, normal. Distinct urothelium (U), submucosa (S) and muscularis (M) layers. Submucosa contains occasional resident lymphocytes. HE. $\times 100$. (B) Cat, urinary bladder, suspected FIC. Variably damaged urothelium (U), submucosal blood vessel dilation and increased lymphocyte infiltration. Submucosa (S) expanded by oedema and haemorrhage (OH). HE. $\times 100$.

variable pancystitis, muscularis fibrosis and mastocytosis (Kabjorn-Gustafsson and Peeker, 2013; Kim *et al*, 2017; Akiyama *et al*, 2019), with some differences between the two subtypes (Maeda *et al*, 2015; Kim *et al*, 2017). Patients with HBPS have focal to diffuse urothelial ulceration, severe lymphocytic inflammation throughout the entire bladder wall, including perineural and perivascular infiltrates, and increased numbers of mast cells in the mucosa and muscularis (Fall *et al*, 1987; Kim *et al*, 2017), as well as muscularis fibrosis (Kabjorn-Gustafsson and Peeker, 2013). On the other hand, patients with NHBPS have mostly normal urothelium and only mild mononuclear inflammation, with the primary changes being mucosal disruption, suburothelial haemorrhages and more prominent fibrosis in the interstitium of the muscularis (Kabjorn-Gustafsson and Peeker, 2013; Kim *et al*, 2017; Akiyama *et al*, 2019) (Table 1).

It is usually very difficult to obtain bladder tissue from cats with FIC as the small size of cats limits the use of cystoscopy and cystoscopic biopsy, particularly because of the small urethral size in male cats (Osborne *et al*, 1996a). The histological lesions in FIC cases include damaged urothelium, submucosal oedema, submucosal blood vessel dilation, submucosal haemorrhage, increased submucosal infiltration of mast cells, mild lymphoplasmacytic inflammation in the submucosa, fibrosis and increased sensory nerve fibre density determined by special stains (Buffington *et al*, 1999; Chew *et al*, 2011b). Neutrophilic infiltrates are uncommon. The histological findings of FIC are comparable with those described in NHBPS, although the mononuclear cell inflammation in FIC is often more severe than that seen in NHBPS (Kim *et al*, 2017; Kullmann *et al*, 2018).

FIC and urolithiasis cases were both found to have elevated bladder wall lymphocyte counts compared with normal bladders, and there was a higher proportion of T lymphocytes than B lymphocytes (Specht *et al*, 2004). Increased numbers of light chain restricted B lymphocytes have recently been associated with HBPS (Maeda *et al*, 2015) but are yet to be evaluated in feline bladder disease. Light chain restriction suggests clonal expansion of B lymphocytes, a process typically comprising part of the local immune response but also occurring in autoimmune disorders (Maeda *et al*, 2015). The increased number of T cells compared with B cells in FIC suggests that FIC is less like HBPS, and perhaps more similar to NHBPS. It is important to note that there are no publications on the normal number of resident lymphocytes in the bladder wall of dogs or cats, and very few in humans. In humans, one paper reports normal urothelium and

submucosa as containing up to 42 lymphocytes per field (area unspecified but probably a standard field size at x 150 magnification) (Christmas, 1994). Another paper reported small numbers of lymphocytes in the submucosa of normal human bladder, particularly CD4+ lymphocytes (Gardiner *et al*, 1986). There are no equivalent reports for canine and feline urinary bladders.

The role of mast cells in BPS and FIC has been strongly contested (van de Merwe *et al*, 2008), with some authors believing mastocytosis should not be used as a definitive criterion for the diagnosis of BPS, because it also occurs in other non-specific chronic inflammatory diseases (Lynes *et al*, 1990; Dundore *et al*, 1996; Specht *et al*, 2004; Wyndaele *et al*, 2009; Liu *et al*, 2012). Mastocytosis has been detected in the submucosal layer (Tomaszewski *et al*, 2001; Kabjorn-Gustafsson and Peeker, 2013) of patients with BPS (Feltis *et al*, 1987; Lynes *et al*, 1987; Lowe *et al*, 1997; van de Merwe *et al*, 2008) and also in the muscularis, with six to eight times the numbers found in control patients (Theoharides *et al*, 2001).

It is believed that mast cell degranulation contributes to the inflammation and fibrosis seen in BPS patients (Wilkinson and Erickson, 2006; Erickson *et al*, 2008; Grover *et al*, 2011; Yoshimura *et al*, 2014). Comparably, mast cell degranulation in cats with FIC is believed to cause release of vasoactive and inflammatory mediators that could contribute to increased bladder wall hypersensitivity and pain (Gunn-Moore, 2002; Kullmann *et al*, 2018), and increased numbers of degranulated mast cells have been observed in the submucosa of FIC patients (Kullmann *et al*, 2018). Conversely, there are reports of similar numbers of submucosal mast cells in BPS and controls (Lynes *et al*, 1987; Dundore *et al*, 1996). The contradictory information on mast cells in BPS and FIC is consistent with the multifactorial nature and variability of these conditions.

Von Brunn's Nests

Von Brunn's nests are invaginations of hyperplastic urothelium that appear continuous with, or immediately subjacent to, the urothelium (Young, 2009), and have been associated with irritative bladder stimuli in animals (Militerno *et al*, 2016). Histologically, von Brunn's nests are foci of normal urothelial cell islands in the submucosa, sometimes with a central cystic lumen, and are suggested to be secondary to chronic urothelial irritation (Militerno *et al*, 2016). In the human literature, there has been debate about whether these nests represent pre-neoplastic change (Weiner *et al*, 1979; Meuten and Meuten, 2016). Von Brunn's

nest cells immunolabel for p27, a cell-cycle inhibitor, but lack tumour markers such as p53 (Volmar *et al*, 2003), indicating that they are most likely not early neoplastic lesions. Although von Brunn's nests are common in normal human bladders and are present in up to 90% of autopsied bladders (Clouston and Lawrentschuk, 2013), their presence has not been specifically evaluated in BPS.

Von Brunn's nests are uncommonly reported in cats but have previously been associated with bladder adenocarcinoma (Patnaik *et al*, 1986) and have been found in conjunction with oedema, haemorrhage, angiogenesis and lymphoplasmacytic inflammation in the ureteral submucosa of two cats that had urethral obstruction and possible FIC (Militerno *et al*, 2016). Recent work has identified von Brunn's nests in the submucosa of 38% of FIC bladder tissue compared with 6% of control feline bladder tissue (Kullmann *et al*, 2018), suggesting a possible link between FIC and the presence of these urothelial nests. Von Brunn's nests have also been identified in the proximal urethral epithelium of FIC cats despite the role of this urothelium in maintaining barrier integrity. This finding suggests a more sensory role for this urethral epithelium than previously believed (Kullmann *et al*, 2018). It is believed that von Brunn's nests may form due to increased cyclooxygenase-1 expression in the bladder submucosa, secondary to increased numbers of degranulated mast cells in this region (Kullmann *et al*, 2018).

Conclusions

This review demonstrates the potential value of comparative pathology and translational medicine in the study of BPS and FIC and indicates that further research is required on the precise aetiology of both conditions. Cats are valuable disease models for human diseases as they share a similar environment and are exposed to similar stressors and external stimuli. Compared with rodents, the longer lifespan of cats allows the chronological contraction of age-related disorders and a longer time for disease and treatment monitoring (Jabr and Fry, 2013). FIC is widely supported as a model for BPS, particularly NHBPS (Jabr and Fry, 2013; Hauser *et al*, 2015), as naturally occurring FIC reproduces more comparable bladder disease to BPS than any other inflammatory bladder disease model (Westropp and Buffington, 2002). Ongoing research will continue to explore the pathogenesis of FIC and BPS, in particular the investigation of urine and tissue biomarkers that may elucidate disease pathogenesis, as well as new methods for diagnosis and treatment. However, FIC is not

comparable to HBPS, which appears to be a different disease entity.

CRediT Author Statement

Emily Jones: Investigation, Writing - original draft preparation, Writing - review and editing. Chiara Palmieri: Writing - reviewing and editing. Mary Thompson: Writing - reviewing and editing. Karen Jackson: Writing - reviewing and editing. Rachel Allavena: Conceptualization, Supervision, Writing - reviewing and editing.

Acknowledgments

We wish to acknowledge the David Galbraith Companion Animal Research Scholarship for financial support.

Conflict of Interest Statement

The authors declared no conflict of interest in relation to the research, authorship or publication of this article.

Funding

This work was supported by the John and Mary Kibble Trust and the University of Queensland School of Veterinary Science Internal Donations Fund.

References

- Akiyama Y, Homma Y, Maeda D (2019) Pathology and terminology of interstitial cystitis/bladder pain syndrome: a review. *Histology & Histopathology*, **24**, 25–32.
- Amat M, Camps T, Manteca X (2016) Stress in owned cats: behavioural changes and welfare implications. *Journal of Feline Medicine & Surgery*, **18**, 577–586.
- Arora HC, Shoskes DA (2015) The enigma of men with interstitial cystitis/bladder pain syndrome. *Translational Andrology and Urology*, **4**, 668–676.
- Baskin L, Tanagho E (1992) Pelvic pain without pelvic organs. *The Journal of Urology*, **147**, 683–686.
- Birder LA (2014) Urinary bladder, cystitis and nerve/urothelial interactions. *Autonomic Neuroscience*, **182**, 89–94.
- Birder LA, Barrick SR, Roppolo JR, Kanai AJ, de Groat WC *et al* (2003) Feline interstitial cystitis results in mechanical hypersensitivity and altered ATP release from bladder urothelium. *American Journal of Physiology - Renal Physiology*, **285**, F423–F429.
- Birder LA, Hanna-Mitchell AT, Mayer E, Buffington CA (2011) Cystitis, co-morbid disorders and associated epithelial dysfunction. *Neurourology and Urodynamics*, **30**, 668–672.
- Birder LA, Wolf-Johnston A, Buffington CA, Roppolo JR, de Groat WC *et al* (2005) Altered inducible nitric oxide synthase expression and nitric oxide production in the

- bladder of cats with feline interstitial cystitis. *The Journal of Urology*, **173**, 625–629.
- Buffington CA, Chew DJ, Woodworth BE (1999) Feline interstitial cystitis. *Journal of the American Veterinary Medical Association*, **215**, 682–687.
- Buffington CAT (2004) Comorbidity of interstitial cystitis with other unexplained clinical conditions. *The Journal of Urology*, **172**, 1242–1248.
- Buffington CAT (2009) Developmental influences on medically unexplained symptoms. *Psychotherapy and Psychosomatics*, **78**, 139–144.
- Buffington CAT (2011a) *Pandora Syndrome: Rethinking Our Approach to Idiopathic Cystitis in Cats* DVM360, October 1. <http://veterinary-medicine.dvm360.com/pandora-syndrome-rethinking-our-approach-idiopathic-cystitis-cats>.
- Buffington CAT (2011b) Idiopathic cystitis in domestic cats - beyond the urinary tract. *Journal of Veterinary Internal Medicine*, **25**, 784–796.
- Buffington CAT, Pacak K (2001) Increased plasma norepinephrine concentration in cats with interstitial cystitis. *The Journal of Urology*, **165**, 2051–2054.
- Buffington CAT, Teng B, Somogyi GT (2002) Norepinephrine content and adrenoceptor function in the bladder of cats with feline interstitial cystitis. *The Journal of Urology*, **167**, 1876–1880.
- Buffington CAT, Wolfe SA (1998) High affinity binding sites for [3H] substance P in urinary bladders of cats with interstitial cystitis. *The Journal of Urology*, **160**, 605–611.
- Cameron ME, Casey RA, Bradshaw JWS, Waran NK, Gunn-Moore DA (2004) A study of environmental and behavioural factors that may be associated with feline idiopathic cystitis. *Journal of Small Animal Practice*, **45**, 144–147.
- Chew D, Buffington CAT (2013) Pandora syndrome: it's more than just the bladder. In: *Proceedings of the American Association of Feline Practitioners*, Dallas, pp 75–83.
- Chew DJ, DiBartola SP, Schenck PA (2011a) *Canine and Feline Nephrology and Urology*, 2nd Edit, Saunders, St. Louis.
- Chew DJ, DiBartola SP, Schenck PA (2011b) Nonobstructive idiopathic or interstitial cystitis in cats. In: *Canine and Feline Nephrology and Urology*, 2nd Edit, DJ Chew, SP DiBartola, PA Schenck, Eds, Saunders, St. Louis, pp 306–340.
- Christmas TJ (1994) Lymphocyte sub-populations in the bladder wall in normal bladder, bacterial cystitis and interstitial cystitis. *British Journal of Urology*, **73**, 508–515.
- Clouston D, Lawrentschuk N (2013) Metaplastic conditions of the bladder. *British Journal of Urology International*, **112**(Suppl 2), 27–31.
- Davis NF, Brady CM, Creagh T (2014) Interstitial cystitis/painful bladder syndrome: epidemiology, pathophysiology and evidence-based treatment options. *European Journal of Obstetrics & Gynecology and Reproductive Biology*, **175**, 30–37.
- Defauw P, Van de Maele I, Duchateau L, Daminet S (2010) Evaluation of possible risk factors for feline idiopathic cystitis. In: *Proceedings of the 53rd BSAVA Annual Congress*, Birmingham, pp 478–479.
- Defauw PA, Van de Maele I, Duchateau L, Polis IE, Saunders JH *et al* (2011) Risk factors and clinical presentation of cats with feline idiopathic cystitis. *Journal of Feline Medicine & Surgery*, **13**, 967–975.
- Denson MA, Griebeling TL, Cohen MB, Kreder KJ (2000) Comparison of cystoscopic and histological findings in patients with suspected interstitial cystitis. *The Journal of Urology*, **164**, 1908–1911.
- DiBartola SP (2010) Clinical approach and laboratory evaluation of renal disease. In: *Textbook of Veterinary Internal Medicine*, SJ Ettinger, EC Feldman, Eds, Elsevier Saunders, St. Louis, pp 1955–1969.
- Dinis S, de Oliveira JT, Pinto R, Cruz F, Buffington CT *et al* (2015) From bladder to systemic syndrome: concept and treatment evolution of interstitial cystitis. *International Journal of Women's Health*, **7**, 735–744.
- Dorsch R, Remer C, Sauter-Louis C, Hartmann K (2014) Feline lower urinary tract disease in a German cat population. *Tierarztl Prax Ausg K Kleintiere Heimtiere*, **4**, 1–9.
- Dundore PA, Schwartz AM, Semerjian H (1996) Mast cell counts are not useful in the diagnosis of nonulcerative interstitial cystitis. *The Journal of Urology*, **155**, 885–887.
- Dyer AJ, Twiss CO (2014) Painful bladder syndrome: an update and review of current management strategies. *Current Urology Reports*, **15**, 384.
- Erickson DR, Tomaszewski JE, Kunselman AR, Stetter CM, Peters KM *et al* (2008) Urine markers do not predict biopsy findings or presence of bladder ulcers in interstitial cystitis/painful bladder syndrome. *The Journal of Urology*, **179**, 1850–1856.
- Fall M, Johansson SL, Aldenborg F (1987) Chronic interstitial cystitis: a heterogeneous syndrome. *The Journal of Urology*, **137**, 35–38.
- Fariello JY, Whitmore KE (2013) Clinical evaluation and diagnosis of bladder pain syndrome. In: *Bladder Pain Syndrome - A Book for Clinicians*, J Nordling, JJ Wyndaele, JP Van de Merwe, P Bouchelouche, M Cervigni, M Fall, Eds, Springer, New York, pp 189–203.
- Feltis JT, Perez-Marrero R, Emerson LE (1987) Increased mast cells of the bladder in suspected cases of interstitial cystitis: a possible disease marker. *The Journal of Urology*, **138**, 42–43.
- Fleshner M, Frank M, Maier SF (2017) Danger signals and inflammasomes: stress-evoked sterile inflammation in mood disorders. *Neuropsychopharmacology*, **42**, 36–45.
- Gardiner RA, Seymore GJ, Lavin MF, Strutton GM, Gemmell E *et al* (1986) Immunohistochemical analysis of the human bladder. *British Journal of Urology*, **58**, 19–25.
- Gerber B, Eichenberger S, Reusch CE (2008) Guarded long-term prognosis in male cats with urethral obstruction. *Journal of Feline Medicine & Surgery*, **10**, 16–23.
- Grover S, Srivastava A, Lee R, Tewari AK, Te AE (2011) Role of inflammation in bladder function and interstitial cystitis. *Therapeutic Advances in Urology*, **3**, 19–33.

- Gunn-Moore DA (2002) *Feline Lower Urinary Tract Disease*. ESFM Feline Congress, Stockholm, pp 133–138.
- Hague DW, Stella JL, Buffington CAT (2013) Effects of interstitial cystitis on the acoustic startle reflex in cats. *American Journal of Veterinary Research*, **74**, 144–147.
- Hauser PJ, VanGordon SB, Seavey J, Sofinowski TM, Ramadan M *et al* (2015) Abnormalities in expression of structural, barrier and differentiation related proteins, and chondroitin sulfate in feline and human interstitial cystitis. *The Journal of Urology*, **194**, 571–577.
- Houbrechts M (2018) *Predisposing Factors of Feline Idiopathic Cystitis in The Netherlands*. Masters Thesis. Utrecht University.
- Hunner GL (1915) A rare type of bladder ulcer in women; report of cases. *Boston Medical & Surgical Journal*, **172**, 660–664.
- Ikeda Y, Birder L, Buffington C, Roppolo J, Kanai A (2009) Mucosal muscarinic receptors enhance bladder activity in cats with feline interstitial cystitis. *The Journal of Urology*, **181**, 1415–1422.
- Izgi K, Daneshgari F (2013) Animal models for interstitial cystitis/painful bladder syndrome (IC/PBS). *Erciyes Tıp Dergisi/Erciyes Medical Journal*, **35**, 103–107.
- Jabr RI, Fry CH (2013) Animal models of lower urinary tract dysfunction. In: *Animal Models for the Study of Human Disease*, M Conn, Ed, Elsevier, St. Louis, pp 461–481.
- Janicki T (2003) Chronic pelvic pain as a form of complex regional pain complex syndrome. *Clinical Obstetrics and Gynecology*, **46**, 797–803.
- Jhang JF, Kuo HC (2016) Pathomechanism of interstitial cystitis/bladder pain syndrome and mapping the heterogeneity of disease. *International Neurourology Journal*, **20**, 95–104.
- Kabjorn-Gustafsson C, Pecker R (2013) Biopsy retrieval, tissue handling, morphology, and histopathological characteristics. In: *Bladder Pain Syndrome: A Guide for Clinicians*, J Nordling, JJ Wyndaele, JP Van de Merwe, P Bouchelouche, M Cervigni, M Fall, Eds, Springer, New York, pp 231–239.
- Keller JJ, Chen YK, Lin HC (2012) Comorbidities of bladder pain syndrome/interstitial cystitis: a population-based study. *British Journal Of Urology International*, **110**, E903–E909.
- Kennedy SM, Lulich JP, Ritt MG, Furrow E (2016) Comparison of body condition score and urinalysis variables between dogs with and without calcium oxalate uroliths. *Journal of the American Veterinary Medical Association*, **249**, 1274–1280.
- Kim A, Han JY, Ryu CM, Yu HY, Lee S *et al* (2017) Histopathological characteristics of interstitial cystitis/bladder pain syndrome without Hunner lesion. *Histopathology*, **71**, 415–424.
- Kirk CA, Lund E, Armstrong P (2001) Prevalence of lower urinary tract disorders of dogs and cats in the United States. In: *Waltham International Symposium Conference Proceedings, Vancouver*, p 61.
- Kiuchi H, Tsujimura A, Takao T, Yamamoto K, Nakayama J *et al* (2009) Increased vascular endothelial growth factor expression in patients with bladder pain syndrome/interstitial cystitis: its association with pain severity and glomerulations. *British Journal Of Urology International*, **104**, 826–831.
- Kruger J, Osborne C, Goyal S, Wickstrom S, Johnston G *et al* (1991) Clinical evaluation of cats with lower urinary tract disease. *Journal of the American Veterinary Medical Association*, **199**, 211–216.
- Kruger JM, Osborne CA, Lulich JP (2009) Changing paradigms of feline idiopathic cystitis. *Veterinary Clinics of North America: Small Animal Practice*, **39**, 15–40.
- Kullmann FA, McDonnell BM, Wolf-Johnston AS, Lynn AM, Giglio D *et al* (2018) Inflammation and tissue remodeling in the bladder and urethra in feline interstitial cystitis. *Frontiers in Systems Neuroscience*, **12**, 1–15.
- Lascelles BD, Knazovicky D, Case B, Freire M, Innes JF *et al* (2015) A canine-specific anti-nerve growth factor antibody alleviates pain and improves mobility and function in dogs with degenerative joint disease-associated pain. *BMC Veterinary Research*, **11**, 101.
- Leiby BE, Landis JR, Probert KJ, Tomaszewski JE, Interstitial Cystitis Data Base Study Group (2007) Discovery of morphological subgroups that correlate with severity of symptoms in interstitial cystitis: a proposed biopsy classification system. *The Journal of Urology*, **177**, 142–148.
- Lekcharoensuk C, Osborne CA, Lulich JP (2001) Epidemiologic study of risk factors for lower urinary tract diseases in cats. *Journal of the American Veterinary Medical Association*, **218**, 1429–1435.
- Lemberger SI, Deeg CA, Hauck SM, Amann B, Hirmer S *et al* (2011) Comparison of urine protein profiles in cats without urinary tract disease and cats with idiopathic cystitis, bacterial urinary tract infection, or urolithiasis. *American Journal of Veterinary Research*, **72**, 1407–1415.
- Liu HT, Shie JH, Chen SH, Wang YS, Kuo HC (2012) Differences in mast cell infiltration, E-cadherin, and zonula occludens-1 expression between patients with overactive bladder and interstitial cystitis/bladder pain syndrome. *Urology*, **80** 225, e213–228.
- Logadottir YR, Ehren I, Fall M, Wiklund NP, Pecker R *et al* (2004) Intravesical nitric oxide production discriminates between classic and nonulcer interstitial cystitis. *The Journal of Urology*, **171**, 1148–1150, discussion 1150–1151.
- Lokeshwar VB, Selzer MG, Cerwinka WH, Gomez MF, Kester RR *et al* (2005) Urinary uronate and sulfated glycosaminoglycan levels: markers for interstitial cystitis severity. *The Journal of Urology*, **174**, 344–349.
- Lowe EM, Anand P, Terengi G, Williams-Chestnut RE, Sinicropi DV *et al* (1997) Increased nerve growth factor levels in the urinary bladder of women with idiopathic sensory urgency and interstitial cystitis. *British Journal of Urology*, **79**, 572–577.
- Lucon M, Dreyfuss J, Silva A, Leite K, Nader H *et al* (2010) Glycosaminoglycans in bladder biopsies and in urine as

- possible biomarkers for interstitial cystitis/bladder pain syndrome. *Urology*, **76**, S98.
- Lulich J, Osborne C, Kruger J (2010) What constitutes a diagnosis of feline idiopathic cystitis. In: *Proceedings American College of Veterinary Internal Medicine Forum, Anaheim*, pp 630–631.
- Lund EM, Armstrong PJ, Kirk CA (1999) Health status and population characteristics of dogs and cats examined at private veterinary practices in the United States. *Journal of the American Veterinary Medical Association*, **214**, 1336–1341.
- Lund HS, Rimstad E, Eggertsdottir AV (2012) Prevalence of viral infections in Norwegian cats with and without feline lower urinary tract disease. *Journal of Feline Medicine & Surgery*, **14**, 895–899.
- Lynes W, Flynn S, Shortliffe L, Lemmers M, Zipser R *et al* (1987) Mast cell involvement in interstitial cystitis. *The Journal of Urology*, **138**, 746–752.
- Lynes W, Flynn S, Shortliffe L, Stamey T (1990) The histology of interstitial cystitis. *The American Journal of Surgical Pathology*, **14**, 969–976.
- Maeda D, Akiyama Y, Morikawa T, Kunita A, Ota Y *et al* (2015) Hunner-type (classic) interstitial cystitis: a distinct inflammatory disorder characterized by pancystitis, with frequent expansion of clonal B-cells and epithelial denudation. *PloS One*, **10**, e0143316.
- Malykhina A, Hanno P (2014) How are we going to make progress treating bladder pain syndrome? *Neurourology and Urodynamics*, **33**, 625–629.
- Meuten DJ, Meuten TLK (2016) Tumors of the urinary system. In: *Tumors in Domestic Animals*, 5th Edit, DJ Meuten, Ed, John Wiley & Sons, Ames, p 687.
- Militerno G, Bassi P, Bettini G, Nannini R (2016) Von Brunn's nests in the ureter of two cats. In: *4th ESVP Meeting of the European Society of Veterinary Pathology – 27th ECVF Meeting of the European College of Veterinary Pathologists, Bologna*.
- Mohamaden WI, Hamad R, Bahr HI (2019) Alterations of pro-inflammatory cytokines and tissue protein expressions in cats with interstitial cystitis. *Pakistan Veterinary Journal*, **39**, 151–156.
- Moldwin RM (2002) Similarities between interstitial cystitis and male chronic pelvic pain syndrome. *Current Urology Reports*, **3**, 313–318.
- Nicolle LE (2006) Asymptomatic bacteriuria: review and discussion of the IDSA guidelines. *International Journal of Antimicrobial Agents*, **28**(Suppl 1), S42–S48.
- Okafor CC, Pearl DL, Blois SL, Lefebvre SL, Yang M *et al* (2019) Factors associated with hematuric struvite crystalluria in cats. *Journal of Feline Medicine & Surgery*, **21**, 922–930.
- Osborne CA, Caywood DD, Johnston GR, Polzin DJ, Lulich JP *et al* (1996a) Feline perineal urethrostomy, a potential cause of feline lower urinary tract disease. *Veterinary Clinics of North America: Small Animal Practice*, **26**, 535–549.
- Osborne CA, Kruger JM, Lulich JP (1996b) Feline lower urinary tract disorders definition of terms and concepts. *Veterinary Clinics of North America: Small Animal Practice*, **26**, 169–179.
- Osborne CA, Lulich JP, Kruger JM, Ulrich LK, Koehler LA (2009) Analysis of 451,891 canine uroliths, feline uroliths, and feline urethral plugs from 1981 to 2007: perspectives from the Minnesota Urolith Center. *Veterinary Clinics of North America: Small Animal Practice*, **39**, 183–197.
- Paape D, Verjans G, Duchateau L, Piron K, Ghys L *et al* (2013) Routine health screening: findings in apparently healthy middle-aged and old cats. *Journal of Feline Medicine & Surgery*, **15**, 8–19.
- Patnaik AK, Schwarz PD, Greene RW (1986) A histopathologic study of twenty urinary bladder neoplasms in the cat. *Journal of Small Animal Practice*, **27**, 433–445.
- Pecker R, Enerback L, Fall M, Aldenborg F (2000) Recruitment, distribution and phenotypes of mast cells in interstitial cystitis. *The Journal of Urology*, **163**, 1009–1015.
- Peters KM, Carrico DJ (2006) Frequency, urgency and pelvic pain: treating the pelvic floor versus the epithelium. *Current Urology Reports*, **7**, 450–455.
- Peters KM, Killinger KA, Mounayer MH, Boura JA (2011) Are ulcerative and nonulcerative interstitial cystitis/painful bladder syndrome 2 distinct diseases? A study of coexisting conditions. *Urology*, **78**, 301–308.
- Rosenberg MT, Hazzard M (2005) Prevalence of interstitial cystitis symptoms in women: a population based study in the primary care office. *The Journal of Urology*, **174**, 2231–2234.
- Roth L, Walton AM, Leib MS, Burrows CF (1990) A grading system for lymphocytic plasmacytic colitis in dogs. *Journal of Veterinary Diagnostic Investigation*, **2**, 257–262.
- Sant GR, Theoharides TC (1994) The role of the mast cell in interstitial cystitis. *Urologic Clinics of North America*, **21**, 41–53.
- Sculptoreanu A, de Groat WC, Buffington CA, Birder LA (2005) Abnormal excitability in capsaicin-responsive DRG neurons from cats with feline interstitial cystitis. *Experimental Neurology*, **193**, 437–443.
- Specht A, Kruger JM, Fitzgerald SD, Kiupel M, Hauptman J (2004) Histochemical and immunohistochemical light microscopic features of chronic feline idiopathic cystitis. In: *Proceedings 22nd Annual American College of Veterinary Internal Medicine Forum, Lakewood, June 9–12*.
- Srikousthubha S, Raghuveer CV, Hingle S (2013) Profile of lesions in cystoscopic bladder biopsies: a histopathological study. *Journal of Clinical and Diagnostic Research*, **7**, 1609–1612.
- Theoharides T, Kempuraj D, Sant G (2001) Mast cell involvement in interstitial cystitis: a review of human and experimental evidence. *Urology*, **57**, 47–55.
- Tirlapur SA, Kuhrt K, Chaliha C, Ball E, Meads C *et al* (2013) The 'evil twin syndrome' in chronic pelvic pain: a systematic review of prevalence studies of bladder pain syndrome and endometriosis. *International Journal of Surgery*, **11**, 233–237.

- Tirumuru S, Al-Kurdi D, Latthe P (2010) Intravesical botulinum toxin A injections in the treatment of painful bladder syndrome/interstitial cystitis: a systematic review. *International Urogynecology Journal*, **21**, 1285–1300.
- Tomaszewski JE, Landis JR, Russack V, Williams TM, Wang L *et al* (2001) Biopsy features are associated with primary symptoms in interstitial cystitis: results from the interstitial cystitis database study. *Urology*, **57**, 67–81.
- van de Merwe J, Arendsen H (2000) Interstitial cystitis: a review of immunological aspects of the aetiology and pathogenesis, with a hypothesis. *British Journal Of Urology International*, **85**, 995–999.
- van de Merwe JP, Nordling J, Bouchelouche P, Bouchelouche K, Cervigni M *et al* (2008) Diagnostic criteria, classification, and nomenclature for painful bladder syndrome/interstitial cystitis: an ESSIC proposal. *European Urology*, **53**, 60–67.
- Volmar KE, Chan TY, De Marzo AM, Epstein JI (2003) Florid von Brunn nests mimicking urothelial carcinoma: a morphologic and immunohistochemical comparison to the nested variant of urothelial carcinoma. *The American Journal of Surgical Pathology*, **27**, 1243–1252.
- Warren JW (2014) Bladder pain syndrome/interstitial cystitis as a functional somatic syndrome. *Journal of Psychosomatic Research*, **77**, 510–515.
- Warren JW, Jackson TL, Langenberg P, Meyers DJ, Xu J (2004) Prevalence of interstitial cystitis in first-degree relatives of patients with interstitial cystitis. *Urology*, **63**, 17–21.
- Warren JW, van de Merwe JP, Nickel JC (2011) Interstitial cystitis/bladder pain syndrome and nonbladder syndromes: facts and hypotheses. *Urology*, **78**, 727–732.
- Weiner DP, Koss LG, Sablay B, Freed SZ (1979) The prevalence and significance of Brunn's nests, cystitis cystica and squamous metaplasia in normal bladders. *The Journal of Urology*, **122**, 317–321.
- Wennevik GE, Meijlink JM, Hanno P, Nordling J (2016) The role of glomerulations in bladder pain syndrome: a review. *The Journal of Urology*, **195**, 19–25.
- Westropp JL (2003) Evaluation of the effects of stress on the sympathetic nervous system and hypothalamic-pituitary-adrenal axis in cats with feline interstitial cystitis. In: *Veterinary Clinical Sciences Graduate Program*, Doctor of Philosophy, Ohio State University, p 153. Available at: https://etd.ohiolink.edu/apexprod/rws_olink/r/1501/6.
- Westropp JL, Buffington CAT (2002) In vivo models of interstitial cystitis. *The Journal of Urology*, **167**, 694–702.
- Westropp JL, Buffington CAT, Chew DJ (2005) Feline lower urinary tract disease. In: *Textbook of Veterinary Internal Medicine*, SJ Ettinger, EC Feldman, Eds, Elsevier Saunders, St. Louis, pp 1828–1850.
- Westropp JL, Welk KA, Buffington CA (2003) Small adrenal glands in cats with feline interstitial cystitis. *The Journal of Urology*, **170**, 2494–2497.
- Whitmore KE, Fall M, Sengiku A, Tomoe H, Logadottir Y *et al* (2019) Hunner lesion versus non-Hunner lesion interstitial cystitis/bladder pain syndrome. *International Journal of Urology*, **26**(Suppl 1), 26–34.
- Wilkinson DR, Erickson DR (2006) Urinary and serologic markers for interstitial cystitis: an update. *Current Urology Reports*, **7**, 414–422.
- Wyndaele JJ, Van Dyck J, Toussaint N (2009) Cystoscopy and bladder biopsies in patients with bladder pain syndrome carried out following ESSIC guidelines. *Scandinavian Journal of Urology and Nephrology*, **43**, 471–475.
- Yoshimura N, Oguchi T, Yokoyama H, Funahashi Y, Yoshikawa S *et al* (2014) Bladder afferent hyperexcitability in bladder pain syndrome/interstitial cystitis. *International Journal of Urology*, **21**(Suppl 1), 18–25.
- Young RH (2009) Tumor-like lesions of the urinary bladder. *Modern Pathology*, **22**(Suppl 2), S37–S52.

[Received, June 15th, 2020
Accepted, March 10th, 2021]